

CSA1711 - Artificial Intelligence

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Analytical Problems with Step-by-Step Solutions

Problem 1: Entropy and Information Gain (Decision Tree)

Given

A dataset has 14 examples for the target attribute 'Play Tennis': 9 examples are 'Yes' and 5 are 'No'. For the attribute 'Wind', the split is: Weak $\rightarrow$ 6 Yes, 2 No (8 total); Strong $\rightarrow$ 3 Yes, 3 No (6 total). Calculate the Information Gain of splitting on 'Wind'.

Step-by-Step Solution

- Step 1. Compute the entropy of the full dataset: $\text{Entropy}(S) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14) = -(0.643)(-0.637) - (0.357)(-1.485) = 0.410 + 0.530 = 0.940$.
- Step 2. Compute the entropy of the 'Weak' subset: $\text{Entropy}(\text{Weak}) = -(6/8)\log_2(6/8) - (2/8)\log_2(2/8) = -(0.75)(-0.415) - (0.25)(-2.0) = 0.311 + 0.500 = 0.811$.
- Step 3. Compute the entropy of the 'Strong' subset: $\text{Entropy}(\text{Strong}) = -(3/6)\log_2(3/6) - (3/6)\log_2(3/6) = -(0.5)(-1.0) - (0.5)(-1.0) = 0.5 + 0.5 = 1.000$.
- Step 4. Compute the weighted average entropy after the split: $\text{Weighted Entropy} = (8/14)(0.811) + (6/14)(1.000) = 0.463 + 0.429 = 0.892$.
- Step 5. Compute the Information Gain: $\text{IG}(S, \text{Wind}) = \text{Entropy}(S) - \text{Weighted Entropy} = 0.940 - 0.892 = 0.048$.

Final Answer

Information Gain for splitting on 'Wind' ≈ 0.048 bits. This low gain indicates 'Wind' is a relatively weak splitting attribute compared to alternatives with higher gain (e.g., 'Outlook', typically ≈ 0.246 in this classic dataset).

Problem 2: Perceptron Weight Update

Given

A perceptron has weights $w = [0.2, -0.4]$ and bias $b = 0.1$. Input $x = [1, 2]$, target output $t = 1$, learning rate $\eta = 0.5$. The activation function is a step function (output 1 if $\text{net} \geq 0$, else 0). Perform one perceptron weight update.

Step-by-Step Solution

- Step 6. Compute the net input: $\text{net} = w \cdot x + b = (0.2 \times 1) + (-0.4 \times 2) + 0.1 = 0.2 - 0.8 + 0.1 = -0.5$.
- Step 7. Apply the step activation function: since $\text{net} = -0.5 < 0$, output $y = 0$.
- Step 8. Compute the error: $\text{error} = t - y = 1 - 0 = 1$.
- Step 9. Update each weight using the perceptron rule: $w_{i_new} = w_i + \eta \times \text{error} \times x_i$. For w_1 : $0.2 + 0.5 \times 1 \times 1 = 0.7$. For w_2 : $-0.4 + 0.5 \times 1 \times 2 = 0.6$.
- Step 10. Update the bias: $b_new = b + \eta \times \text{error} = 0.1 + 0.5 \times 1 = 0.6$.

Final Answer

Updated weights: $w = [0.7, 0.6]$, $b = 0.6$. On the next forward pass, $\text{net} = (0.7 \times 1) + (0.6 \times 2) + 0.6 = 2.5 \geq 0$, so the perceptron now correctly outputs 1.

Problem 3: A* Search — Path Cost Calculation

Given

A search graph has nodes A, B, C, Goal. Edge costs: $A \rightarrow B = 2$, $A \rightarrow C = 3$, $B \rightarrow \text{Goal} = 4$, $C \rightarrow \text{Goal} = 2$. Heuristic estimates (straight-line to Goal): $h(A) = 5$, $h(B) = 3$, $h(C) = 2$, $h(\text{Goal}) = 0$. Use A* search ($f = g + h$) to find the optimal path from A to Goal.

Step-by-Step Solution

Step 11. Evaluate the start node A: $g(A) = 0$, $h(A) = 5$, so $f(A) = 0 + 5 = 5$.

Step 12. Expand A, generating B and C. Compute $f(B) = g(B) + h(B) = 2 + 3 = 5$. Compute $f(C) = g(C) + h(C) = 3 + 2 = 5$.

Step 13. Both B and C tie at $f = 5$; expand the one found first by tie-breaking convention — expand B: from B, reach Goal with $g(\text{Goal via B}) = 2 + 4 = 6$, $f = 6 + 0 = 6$.

Step 14. Expand C: from C, reach Goal with $g(\text{Goal via C}) = 3 + 2 = 5$, $f = 5 + 0 = 5$.

Step 15. Compare the two paths to Goal: path $A \rightarrow B \rightarrow \text{Goal}$ has $f = 6$; path $A \rightarrow C \rightarrow \text{Goal}$ has $f = 5$. A* selects the lower-cost path.

Final Answer

Optimal path: $A \rightarrow C \rightarrow \text{Goal}$, with total path cost $g = 5$, which is lower than the $A \rightarrow B \rightarrow \text{Goal}$ path (cost 6).

Problem 4: Naive Bayes Classification

Given

Given $P(\text{Play}=\text{Yes}) = 0.64$, $P(\text{Play}=\text{No}) = 0.36$, and for a new instance with $\text{Outlook}=\text{Sunny}$, $\text{Wind}=\text{Strong}$: $P(\text{Sunny}|\text{Yes}) = 0.22$, $P(\text{Strong}|\text{Yes}) = 0.33$, $P(\text{Sunny}|\text{No}) = 0.60$, $P(\text{Strong}|\text{No}) = 0.60$. Classify the instance using Naive Bayes.

Step-by-Step Solution

Step 16. Write the Naive Bayes formula: $P(\text{Class}|\text{features}) \propto P(\text{Class}) \times \prod P(\text{feature}_i|\text{Class})$.

Step 17. Compute the unnormalized score for Yes: $P(\text{Yes}) \times P(\text{Sunny}|\text{Yes}) \times P(\text{Strong}|\text{Yes}) = 0.64 \times 0.22 \times 0.33 = 0.0465$.

Step 18. Compute the unnormalized score for No: $P(\text{No}) \times P(\text{Sunny}|\text{No}) \times P(\text{Strong}|\text{No}) = 0.36 \times 0.60 \times 0.60 = 0.1296$.

Step 19. Normalize (optional, for probability output): $\text{total} = 0.0465 + 0.1296 = 0.1761$. $P(\text{Yes}|\text{features}) = 0.0465/0.1761 = 0.264$. $P(\text{No}|\text{features}) = 0.1296/0.1761 = 0.736$.

Step 20. Compare the two class scores and select the larger one.

Final Answer

Since the score for 'No' (0.1296) is greater than 'Yes' (0.0465), the Naive Bayes classifier predicts Play = No, with approximately 73.6% posterior probability.

Problem 5: K-Means Clustering — One Iteration

Given

Points: A(2,2), B(3,3), C(8,8), D(9,9). Initial centroids: $C1 = (2,2)$, $C2 = (9,9)$. Perform one iteration of K-Means (assign points, then recompute centroids) using Euclidean distance.

Step-by-Step Solution

- Step 21. Compute distance of each point to C1(2,2) and C2(9,9). For A(2,2): $d(A,C1)=0$, $d(A,C2)=\sqrt{((9-2)^2+(9-2)^2)}=\sqrt{98}\approx 9.90$.
- Step 22. For B(3,3): $d(B,C1)=\sqrt{(1^2+1^2)}=\sqrt{2}\approx 1.41$, $d(B,C2)=\sqrt{(6^2+6^2)}=\sqrt{72}\approx 8.49$.
- Step 23. For C(8,8): $d(C,C1)=\sqrt{(6^2+6^2)}=\sqrt{72}\approx 8.49$, $d(C,C2)=\sqrt{(1^2+1^2)}=\sqrt{2}\approx 1.41$.
- Step 24. For D(9,9): $d(D,C1)=\sqrt{98}\approx 9.90$, $d(D,C2)=0$. Assign each point to its nearer centroid: Cluster1 = {A, B}, Cluster2 = {C, D}.
- Step 25. Recompute centroids as the mean of each cluster: New C1 = $((2+3)/2, (2+3)/2) = (2.5, 2.5)$. New C2 = $((8+9)/2, (8+9)/2) = (8.5, 8.5)$.

Final Answer

After one iteration: Cluster1 = {A, B} with centroid (2.5, 2.5); Cluster2 = {C, D} with centroid (8.5, 8.5). Since assignments are stable and well-separated, a second iteration would produce the same clusters — the algorithm has converged.

Problem 6: Minimax Algorithm — Game Tree Evaluation

Given

A game tree has a MAX root node with two MIN children, each MIN node having two leaf nodes. Leaf values (left to right): 3, 5, 2, 9. Compute the Minimax value of the root.

Step-by-Step Solution

- Step 26. Identify tree structure: Root (MAX) → MIN-Left, MIN-Right. MIN-Left → leaves [3, 5]. MIN-Right → leaves [2, 9].
- Step 27. Evaluate MIN-Left: MIN node selects the minimum of its children's values: $\min(3, 5) = 3$.
- Step 28. Evaluate MIN-Right: $\min(2, 9) = 2$.
- Step 29. Evaluate the Root (MAX node): MAX selects the maximum of its children's values: $\max(\text{MIN-Left}, \text{MIN-Right}) = \max(3, 2) = 3$.
- Step 30. Trace back the optimal path: the MAX player should choose the branch leading to MIN-Left, since it guarantees a value of 3 (better than the 2 guaranteed by the other branch).

Final Answer

The Minimax value of the root is 3, and the optimal move for the MAX player is to go toward the MIN-Left subtree.

Problem 7: Gradient Descent — Single Weight Update

Given

A single-neuron linear model predicts $\hat{y} = w \cdot x$, with current $w = 2$, input $x = 3$, true target $y = 10$, and learning rate $\eta = 0.01$. Loss function is Mean Squared Error: $L = (y - \hat{y})^2$. Perform one gradient descent update on w .

Step-by-Step Solution

- Step 31. Compute the prediction: $\hat{y} = w \times x = 2 \times 3 = 6$.
- Step 32. Compute the loss: $L = (y - \hat{y})^2 = (10 - 6)^2 = 16$.
- Step 33. Compute the gradient of L with respect to w : $dL/dw = 2(\hat{y} - y) \times x = 2(6 - 10) \times 3 = 2 \times (-4) \times 3 = -24$.
- Step 34. Apply the gradient descent update rule: $w_{\text{new}} = w - \eta \times (dL/dw) = 2 - 0.01 \times (-24) = 2 + 0.24 = 2.24$.
- Step 35. Verify the update reduces loss: with $w = 2.24$, new $\hat{y} = 2.24 \times 3 = 6.72$, which is closer to the target 10 than the previous prediction of 6.

Final Answer

Updated weight $w = 2.24$. The prediction moved from 6 to 6.72, confirming the gradient step reduced the loss in the correct direction.

Problem 8: K-Nearest Neighbors (KNN) Classification

Given

Training points with labels: $P1(1,1)=A$, $P2(2,1)=A$, $P3(4,4)=B$, $P4(5,5)=B$. Classify a new point $Q(3,2)$ using KNN with $k=3$ and Euclidean distance.

Step-by-Step Solution

Step 36. Compute distance from $Q(3,2)$ to each training point. $d(Q,P1) = \sqrt{((3-1)^2+(2-1)^2)} = \sqrt{4+1} = \sqrt{5} \approx 2.24$.

Step 37. $d(Q,P2) = \sqrt{((3-2)^2+(2-1)^2)} = \sqrt{1+1} = \sqrt{2} \approx 1.41$.

Step 38. $d(Q,P3) = \sqrt{((3-4)^2+(2-4)^2)} = \sqrt{1+4} = \sqrt{5} \approx 2.24$.

Step 39. $d(Q,P4) = \sqrt{((3-5)^2+(2-5)^2)} = \sqrt{4+9} = \sqrt{13} \approx 3.61$.

Step 40. Rank distances and select the $k=3$ nearest neighbors: $P2(1.41, A)$, then $P1$ and $P3$ tied at 2.24 (A and B). The 3 nearest are $P2(A)$, $P1(A)$, $P3(B)$ — majority label among these three is A (2 votes vs. 1).

Final Answer

$Q(3,2)$ is classified as class A , since 2 of its 3 nearest neighbors ($P1$ and $P2$) belong to class A .

Problem 9: Markov Decision Process — Bellman Update

Given

A state s has two possible actions. Action 1 leads to a reward of 5 and next-state value $V(s') = 10$. Action 2 leads to a reward of 8 and next-state value $V(s') = 4$. Discount factor $\gamma = 0.9$. Compute $V(s)$ using one step of value iteration (Bellman optimality equation).

Step-by-Step Solution

Step 41. Write the Bellman optimality equation: $V(s) = \max \text{ over actions of } [R(s,a) + \gamma \times V(s')]$.

Step 42. Compute the value for Action 1: $Q1 = R + \gamma V(s') = 5 + 0.9 \times 10 = 5 + 9 = 14$.

Step 43. Compute the value for Action 2: $Q2 = R + \gamma V(s') = 8 + 0.9 \times 4 = 8 + 3.6 = 11.6$.

Step 44. Compare the two action values: $Q1 = 14$ versus $Q2 = 11.6$.

Step 45. Select the maximum, since value iteration takes the best achievable action value: $V(s) = \max(14, 11.6) = 14$.

Final Answer

$V(s) = 14$, achieved by choosing Action 1. This also identifies the optimal policy at state s as Action 1.

Problem 10: Confusion Matrix — Precision, Recall, F1-Score

Given

A binary classifier produces the following results on a test set: True Positives (TP) = 40, False Positives (FP) = 10, False Negatives (FN) = 5, True Negatives (TN) = 45. Compute Accuracy, Precision, Recall, and F1-Score.

Step-by-Step Solution

Step 46. Compute Accuracy: $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) = (40 + 45) / (40 + 45 + 10 + 5) = 85 / 100 = 0.85$.

Step 47. Compute Precision: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 40 / (40 + 10) = 40 / 50 = 0.80$.

Step 48. Compute Recall: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 40 / (40 + 5) = 40 / 45 \approx 0.889$.

Step 49. Compute F1-Score using the harmonic mean of Precision and Recall: $\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 2 \times (0.80 \times 0.889) / (0.80 + 0.889)$.

Step 50. Finish the F1 calculation: numerator = $2 \times 0.711 = 1.422$; denominator = 1.689; $\text{F1} = 1.422 / 1.689 \approx 0.842$.

Final Answer

Accuracy = 0.85 (85%), Precision = 0.80 (80%), Recall $\approx$ 0.889 (88.9%), F1-Score $\approx$ 0.842 (84.2%). The model has higher recall than precision, meaning it misses fewer true positives but generates a moderate number of false positives.